



## Solving a system using Gaussian Elimination & Back Substitution

Let us consider the following system of linear equations:

$$\begin{aligned}x + 2y + z &= 2 \\ 3x + 8y + z &= 12 \\ 4y + z &= 2\end{aligned}$$

We will find the solutions by simplifying the above system through row operations. We can consider the following techniques of row operations:

1. Multiply one equation by a nonzero constant
2. Add one equation to another.

We will describe a technique that will help us to perform row operations more systematically and with greater clarity.

This particular technique is known as ***Gaussian Elimination method***, which can also be stated as ***row echelon form***.

The ***Augmented Matrix*** of the above linear system is denoted by:

$$\left( \begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array} \right).$$

In this case “Augmented” indicates to tag something.

The ***Coefficient Matrix*** of the above system is denoted by:

$$\left( \begin{array}{ccc} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{array} \right).$$

### **Pivot Position:**

A **pivot** position in a **matrix** A is a location in A that corresponds to a leading 1 in the echelon form of A. A **pivot column** is a **column** of A that contains a **pivot** position.

If there is a row of all zeros, then it is at the bottom of the **matrix**. The first non-zero element of any row is a **1**. That element is called the **leading 1**. The **leading 1** of any row is to the right of the **leading 1** of the previous row.

Considering an example:

$$A = \begin{bmatrix} 1 & 4 & 5 & 9 & 7 \\ 0 & 2 & 4 & 6 & 6 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



The matrix A is in echelon form and thus reveals that columns 1, 2, and 5 are pivot columns.

The pivots in this example are 1, 2, and 5.

In the **Gaussian Elimination Method**, the structure of the matrix after elimination should be:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

### **Process:**

1. If  $a_{11} = 0$  then we have to interchange  $R_1$  with any other row.

$$\begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix} R_1, R_2 \text{ interchange}$$

2. If  $a_{11} \neq 0$ , then it is either 1 or any other numerical value known as **Pivot**.

Ex:  $\begin{bmatrix} 1 & 5 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 4 & 0 & 5 \\ -1 & 7 & 3 \\ 3 & 1 & -1 \end{bmatrix}$ .

3. If there exists any row that is entirely zero, should be shifted at the bottom.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ 0 & 0 & 0 \end{bmatrix}$$

4. All the elements under each **pivot** should be 0.

$$\begin{bmatrix} \mathbf{a_{11}} & a_{12} & a_{13} \\ \mathbf{0} & \mathbf{a_{22}} & a_{23} \\ \mathbf{0} & \mathbf{0} & \mathbf{a_{33}} \end{bmatrix}$$

Let us consider the given exam again.

$$\begin{aligned} x + 2y + z &= 2 \\ 3x + 8y + z &= 12 \\ 4y + z &= 2 \end{aligned}$$

We will solve  $Ax = b$ .

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{pmatrix}, x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, b = \begin{pmatrix} 2 \\ 12 \\ 2 \end{pmatrix}.$$

We will start with *Augmented Matrix* of the system:

$$\left( \begin{array}{ccc|c} \mathbf{1} & 2 & 1 & 2 \\ 3 & \mathbf{8} & 1 & 12 \\ 0 & 4 & \mathbf{1} & 2 \end{array} \right)$$

$$= \left( \begin{array}{ccc|c} \mathbf{1} & 2 & 1 & 2 \\ 0 & \mathbf{2} & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array} \right) R_2 \rightarrow R_2 - 3R_1$$

$$= \left( \begin{array}{ccc|c} \mathbf{1} & 2 & 1 & 2 \\ 0 & \mathbf{2} & -2 & 6 \\ 0 & 0 & \mathbf{5} & -10 \end{array} \right) R_3 \rightarrow R_3 - 2R_2$$

Our pivots are 1, 2 and 5.

Now we will solve the system in reverse order since the system is a **triangle**. We call it **Back Substitution**.

Here  $U = \text{upper triangle} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{pmatrix}$  and  $c = \begin{pmatrix} 2 \\ 6 \\ -10 \end{pmatrix}$ .

After the row elimination, the matrix  $A$  becomes  $U$  and  $b$  becomes  $c$ .

Let's write the matrix in terms of system of linear equation:

$$x + 2y + z = 2 \quad (1)$$

$$2y - 2z = 6 \quad (2)$$

$$5z = -10 \quad (3)$$

$$(3) \rightarrow 5z = -10, \therefore z = -2$$

$$(2) \rightarrow 2y - 2(-2) = 6, \therefore y = 1$$

$$(1) \rightarrow x + 2(1) + (-2) = 2, \therefore x = 2$$

$$\therefore \text{Solution of the system } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}.$$

### Row Echelon Form Of Matrix U

$$U = \text{upper triangle} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{pmatrix}$$

$$\text{Row Echelon Form: } \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{matrix} R_2 \rightarrow R_2 \div 2 \\ R_3 \rightarrow R_3 \div 5 \end{matrix}$$

### Elementary row operations & their corresponding Elementary matrices

Consider the upper triangular matrix that we have evaluated after row operation:

$$U = \text{upper triangle} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{pmatrix}. \text{ It contains three pivots: 1, 2 and 5.}$$

The whole purpose of this elimination was to get from  $A$  to  $U$ .

**Elementary Matrix "E":** It is the matrix or product of two or more matrices that takes us from  $A$  to  $U$ .

Consider the following steps:

1. Get a **leading 1** as the first entry of  $R_1$  (Note: Leading 1 can be any constant value except 0)

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}$$

So it is the matrix  $A$  itself.

2. Use the 1<sup>st</sup> Pivot to clear out  $C_1$  (Column 1) as follows: (Everything under the 1<sup>st</sup> Pivot position in Column 1 should be 0)

$$\begin{bmatrix} ? & & \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} R_2 \rightarrow R_2 - 3R_1$$

The corresponding element that changes in the “?” matrix is  $a_{21}$   $\because$  we see the change in  $R_2$  by the row operation  $R_2 - 3R_1$ , while  $R_1$  and  $R_3$  are unchanged in the right hand side.

Note: **Identity matrix** acts like “1” in the **Matrix Multiplication**.

$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ , since in the R.H.S. of step 2,  $R_1$  and  $R_3$  are unchanged and  $R_2$  has been changed by the row operation  $R_2 - 3R_1$  which will affect the element  $a_{21}$  by  $-3$ ,

Hence:

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} R_2 \rightarrow R_2 - 3R_1$$

Let's call  $\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = E_{21}$ , while  $E$  stands for **elementary** or **elimination**.

3. Use the 2<sup>nd</sup> Pivot to clear out  $C_2$  (Column 2) as follows:

$$\begin{bmatrix} ? & & \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} R_3 \rightarrow R_3 - 2R_2 = U$$

The corresponding element that changes in the “?” matrix is  $a_{32}$  ∴ we see the change in  $R_3$  by the row operation  $R_3 - 2R_2$ , while  $R_1$  and  $R_2$  are unchanged in the right hand side.

Since in the R.H.S. of step 3,  $R_1$  and  $R_2$  are unchanged and  $R_3$  has been changed by the row operation  $R_3 - 2R_2$  which will affect the element  $a_{32}$  by  $-2$ ,

**Hence:**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} R_3 \rightarrow R_3 - 2R_2$$

Let's call  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} = E_{32}$

So we have  $E_{32}E_{21}A = U$  therefore  $EA=U$

Therefore in this case Elementary Matrix  $E = E_{32}E_{21}$

$$\begin{aligned} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 6 & -2 & 1 \end{bmatrix} \end{aligned}$$

## The basic concept of Inverse

### Reverse of Elementary Row Operation

- We know that  $EA = U$ , while  $E$  represents elementary matrix.
- We have learnt how to reach  $U$  from  $A$ .
- But the question is how do we get back to  $A$  from  $U$ ?
- ✓ We have to reverse our calculation steps. We can use the word “Inverse” to explain that situation.

Let's think “*what is Inverse?*”

Consider:

$$E = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We want to find a matrix that will undo the elimination and give us *Identity*.

$$\begin{pmatrix} \dots & \dots & \dots \\ \vdots & ? & \vdots \\ \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We have to add 3 times  $R_1$  to  $R_2$  ( $R_2 + 3R_1$ ) to get the Identity. As you can see  $R_1$  and  $R_3$  will remain unchanged.

$$\begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} R_2 \rightarrow R_2 + 3R_1$$

$$\therefore E^{-1}E = I$$

So, basically when we change the signs of an elementary matrix it becomes inverted.

### Inverse of an elementary matrix

Consider the following:

Since  $E_{21}$  is one of the elementary matrices

$$\begin{bmatrix} ? & & \\ & & \\ & & \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The solution is:

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

We kept the signs opposite to each other for the element  $a_{21}$  since the product of these two matrices is  $I_3$ .

$$\therefore E_{21}^{-1}E_{21} = I$$

The matrix  $A$  is invertible if there exists  $A^{-1}$ , that *inverts*  $A$ .

**Two sided inverse:**  $A^{-1}A = I$  &  $AA^{-1} = I$

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow E_{21}^{-1}E_{21} = I$$

**Similarly**

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow E_{21}E_{21}^{-1} = I$$